

Lignite Converted

Data from Handout

In a study concerning coal conservation into oils, the hydrogenolysis of a Canadian lignite using carbon monoxide and hydrogen mixtures as reducing agents was investigated.

:

```
#setwd("F:/Lexar/stat 482 data sets")
lignite.dat = read.csv('lignite.csv')
lignite.dat
```

```
##      temperature molar.ratio pressure time conversion
## 1           380         0.25      7.1  10         53.3
## 2           460         0.25      7.1  10         78.0
## 3           380         0.75      7.1  10         62.4
## 4           460         0.75      7.1  10         78.9
## 5           380         0.25     11.1  10         75.9
## 6           460         0.25     11.1  10         75.4
## 7           380         0.75     11.1  10         71.3
## 8           460         0.75     11.1  10         84.4
## 9           380         0.25      7.1  50         64.5
## 10          460         0.25      7.1  50         67.5
## 11          380         0.75      7.1  50         72.8
## 12          460         0.75      7.1  50         85.3
## 13          380         0.25     11.1  50         71.4
## 14          460         0.25     11.1  50         83.3
## 15          380         0.75     11.1  50         82.9
## 16          460         0.75     11.1  50         81.7
```

```
cor(lignite.dat)
```

```
##           temperature molar.ratio pressure      time conversion
## temperature  1.0000000  0.0000000 0.0000000 0.0000000  0.5735635
## molar.ratio  0.0000000  1.0000000 0.0000000 0.0000000  0.3613450
## pressure     0.0000000  0.0000000 1.0000000 0.0000000  0.4559830
## time         0.0000000  0.0000000 0.0000000 1.0000000  0.2136524
## conversion   0.5735635  0.3613450 0.4559830 0.2136524  1.0000000
```

```
attach(lignite.dat)
```

```
## The following object is masked from package:datasets:
```

```
##
```

```
##      pressure
```

```
lignite.mod = lm(conversion ~ temperature + molar.ratio + pressure + time)
summary(lignite.mod)
```

```
##
```

```
## Call:
```

```
## lm(formula = conversion ~ temperature + molar.ratio + pressure +
```

```
##      time)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -7.0250 -3.5125  0.1375  2.9875  7.6750
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -5.36750    16.55028  -0.324  0.75178
## temperature  0.12500     0.03520   3.552  0.00454 **
## molar.ratio 12.60000     5.63128   2.238  0.04691 *
## pressure     1.98750     0.70391   2.824  0.01656 *
## time         0.09312     0.07039   1.323  0.21269
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 5.631 on 11 degrees of freedom
## Multiple R-squared:  0.7131, Adjusted R-squared:  0.6088
## F-statistic: 6.836 on 4 and 11 DF,  p-value: 0.005123
```

```
anova(lignite.mod)
```

```
## Analysis of Variance Table
##
## Response: conversion
##              Df Sum Sq Mean Sq F value    Pr(>F)
## temperature  1 400.00  400.00 12.6138 0.004539 **
## molar.ratio   1 158.76  158.76  5.0064 0.046906 *
## pressure      1 252.81  252.81  7.9722 0.016563 *
## time          1  55.50   55.50  1.7502 0.212689
## Residuals    11 348.83   31.71
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
x1 = 2*(temperature-mean(temperature))/(range(temperature)[2]-range(temperature)[1])
x2 = 2*(molar.ratio-mean(molar.ratio))/(range(molar.ratio)[2]-range(molar.ratio)[1])
x3 = 2*(pressure-mean(pressure))/(range(pressure)[2]-range(pressure)[1])
x4 = 2*(time-mean(time))/(range(time)[2]-range(time)[1])
```

```
cbind(x1,x2,x3,x4)
```

```
##      x1 x2 x3 x4
## [1,] -1 -1 -1 -1
## [2,]  1 -1 -1 -1
## [3,] -1  1 -1 -1
## [4,]  1  1 -1 -1
## [5,] -1 -1  1 -1
## [6,]  1 -1  1 -1
## [7,] -1  1  1 -1
## [8,]  1  1  1 -1
## [9,] -1 -1 -1  1
## [10,]  1 -1 -1  1
## [11,] -1  1 -1  1
## [12,]  1  1 -1  1
## [13,] -1 -1  1  1
```

```
## [14,] 1 -1 1 1
## [15,] -1 1 1 1
## [16,] 1 1 1 1

orthog.mod = lm(conversion ~ x1+x2+x3+x4)
summary(orthog.mod)

##
## Call:
## lm(formula = conversion ~ x1 + x2 + x3 + x4)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -7.0250 -3.5125  0.1375  2.9875  7.6750
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   74.312      1.408   52.785 1.39e-14 ***
## x1              5.000      1.408    3.552 0.00454 **
## x2              3.150      1.408    2.238 0.04691 *
## x3              3.975      1.408    2.824 0.01656 *
## x4              1.863      1.408    1.323 0.21269
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 5.631 on 11 degrees of freedom
## Multiple R-squared:  0.7131, Adjusted R-squared:  0.6088
## F-statistic: 6.836 on 4 and 11 DF,  p-value: 0.005123

accepted.mod = lm(conversion ~ x1+x2+x3)
summary(accepted.mod)

##
## Call:
## lm(formula = conversion ~ x1 + x2 + x3)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -8.8875 -4.7375  0.8375  4.6750  6.8125
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   74.312      1.451   51.209 2.02e-15 ***
## x1              5.000      1.451    3.446 0.00484 **
## x2              3.150      1.451    2.171 0.05073 .
## x3              3.975      1.451    2.739 0.01796 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 5.805 on 12 degrees of freedom
## Multiple R-squared:  0.6675, Adjusted R-squared:  0.5843
## F-statistic: 8.029 on 3 and 12 DF,  p-value: 0.003352
```